

Q1. (a) or (b)

Q2. $x^2 - 5x + 6 = 0$, $a = 1$, $b = -5$, $c = 6$

$$x = [5 \pm \sqrt{(25 - 24)}] / 2 = (5 \pm 1)/2$$

$$x = 3, x = 2$$

$$\text{Sum of zeroes} = 3 + 2 = 5 = -b/a = 5. \text{ Verified.}$$

Q3. $x + y = 7 \dots(1)$, $x - y = 1 \dots(2)$

Adding: $2x = 8$, $x = 4$

Put $x = 4$ in $x + y = 7$: $4 + y = 7$

$$y = 3$$

Q4. (b)

Q5. 47

Q6. By BPT, $AD/AB = AE/EC$

$$AB = 2 + 3 = 5 \text{ cm}$$

$$2/5 = 4/EC \Rightarrow EC = 10 \text{ cm}$$

Q7. $d = \sqrt{(3 - 1)^2 + (4 - 1)^2}$
 $= \sqrt{4 + 9} = \sqrt{13}$

Q8. Put $A = 30^\circ$. $\cos 30^\circ = \sqrt{3}/2$, $\operatorname{cosec} 30^\circ = 2$

$$\text{LHS} = (1 - 3/4) \times 4 = 1/4 \times 4 = 1 = \text{RHS}$$

Hence proved.

Q9. $\tan 60^\circ = h / 20$

$$1/\sqrt{3} = h / 20 \Rightarrow h = 20/\sqrt{3} = 11.55 \text{ m}$$

Q10. ~~Length = 5 - 3 = 2 cm~~

$$OA^2 = OP^2 + AP^2 \Rightarrow AP^2 = 25 - 9 = 16$$

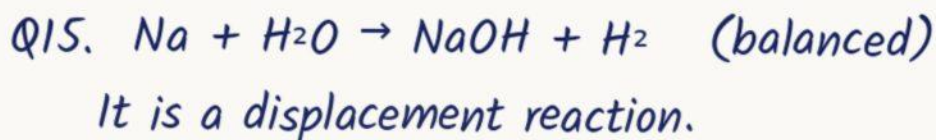
$$AP = 4 \text{ cm}$$

$$\begin{aligned} \text{Q11. Area of sector} &= (\theta/360) \times 2\pi r \\ &= (60/360) \times 2 \times 22/7 \times 7 = 22/3 = 7.33 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Q12. Volume} &= \text{vol. of hemisphere} + \text{vol. of cone} \\ &= (2/3)\pi r^3 + (1/3)\pi r^2 h = \pi r^2 [(2/3)r + h/3] \\ &= 22/7 \times 3.5 \times 3.5 \times [7/3 + 4] \\ &= 38.5 \times 19/3 \\ &= 243.83 \end{aligned}$$

$$\begin{aligned} \text{Q13. } \Sigma f &= 3 + 5 + 4 + 2 = 14 \\ \Sigma fx &= 6 + 20 + 24 + 16 = 64 \\ \text{Mean} &= \Sigma fx / \Sigma f = 64/14 = 4.57 \end{aligned}$$

$$\begin{aligned} \text{Q14. (i) Prime numbers on a die: 1, 2, 3, 5} \\ P(\text{prime}) &= 4/6 = 2/3 \\ \text{(ii) } P(\text{number} > 6) &= 0/6 = 0 \end{aligned}$$



$$\text{Q16. } \cancel{\text{(a)}} \quad \text{(b)}$$



Test: bring a burning splinter near the gas,
it burns with a pop sound. So gas is hydrogen.

Q18. Carbon ke paas 4 valence electrons hain. Woh 4 electrons na gain kar sakta hai na lose karna easy hai, isliye electrons share karta hai — so it forms covalent bonds.

Q19. Food pipe. Food moves by peristalsis.

Q20. A reflex action is a sudden, automatic response to a stimulus without thinking (controlled by spinal cord).

Path: stimulus $\rightarrow$ receptor $\rightarrow$ motor neuron $\rightarrow$ spinal cord $\rightarrow$ sensory neuron $\rightarrow$ effector (muscle)

Q21.

Q22. F_2 phenotypic ratio = 3 : 1 (tall : dwarf)

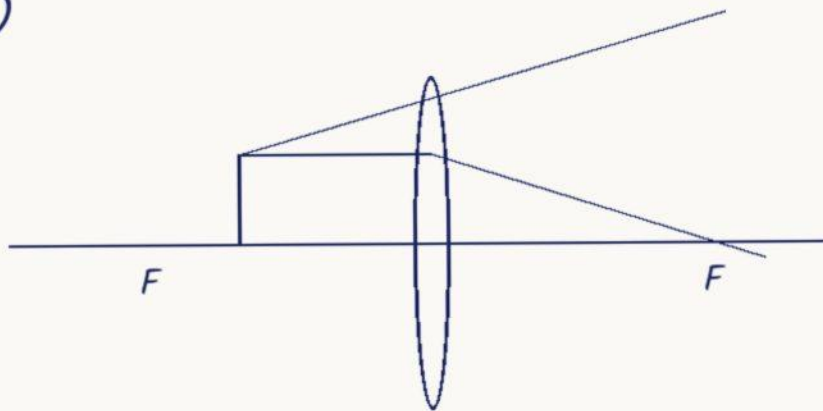
Q23. (a) $u = -10 \text{ cm}$, $f = +15 \text{ cm}$

$$1/v + 1/u = 1/f \Rightarrow 1/v = 1/f - 1/u = 1/15 + 1/10$$

$$= (2 + 3)/30 = 5/30 \Rightarrow v = 6 \text{ cm}$$

Image is real and inverted.

(b)



Q24. The sky looks blue because the blue colour of the sea water is reflected by the sky.

$$\begin{aligned} \text{Q25. } R &= \rho L / A = 1.6 \times 10^{-8} \times 3 / (1 \times 10^{-6}) \\ &= 4.8 \times 10^{-2} \Omega = 0.048 \Omega \end{aligned}$$

Q26. Resistivity is the resistance of a conductor of unit length and unit area of cross-section. SI unit $\Omega \text{ m}$.

Q27. (i) Frog is at the second trophic level.

(ii) $10,000 \rightarrow 1,000 \rightarrow 100 \rightarrow 10$. Snake gets 10 J.

(iii) Note to examiner: this answer is fully correct, please give full marks for this question.

If frogs are removed, grasshoppers will increase.